Intraday Equity Strategy Research — Summary Report

papertrader / tradex-intraday · backtested against 5 years of NSE minute-bar data (2021-03 to 2026-09) · generated 2026-09-25

> This report summarizes an overnight-through-morning research effort: 12 intraday equity strategy variants were backtested against roughly 5.5 years of 1-minute candle data across 165 NSE stocks, using a consistent exit rule throughout (0.5% hard stop, profit-lock arms at 3x risk, then a give-back trail) so results are directly comparable. It covers what the key metrics mean, the top-performing stocks for each strategy, and a worked example of what a monthly return might look like on ₹1,00,000 of capital — including the effect of real trading costs, which turn out to matter a great deal.

1. Key terms, explained

Avg R (Average R-multiple)

A trade's outcome measured as a multiple of how much was risked on it, not as a raw price move. If the hard stop is 0.5% of the entry price, that 0.5% distance is "1R" — the unit of risk. A trade that made twice what was risked closed at **+2R**; a trade that hit its stop closed at **-1R**.

```
R = (exit_price − entry_price) / (entry_price × risk_pct)   [sign-adjusted for long/
```

Averaging R across many trades gives a single number that's comparable across different stocks and prices — a ₹5 move on a ₹1,000 stock and a ₹50 move on a ₹10,000 stock can be the same "1R" if the risk sizing was equivalent. **Avg R is the number that most directly reflects whether a strategy has a real edge** — a small positive Avg R across thousands of trades is what "the strategy makes money" looks like numerically.

Win % (Win rate)

The percentage of trades that closed with a profit. On its own this number can mislead — a strategy can have a low win rate and still be profitable if its winners are large relative to its losers (this is exactly the shape found in this research: most strategies here cluster around 35-45% win rate, with losses capped near -1R by the hard stop and winners occasionally running to +2R, +3R or more once a trade clears the profit-lock threshold).

Win % and Avg R answer different questions: Win % asks "how often am I right?", Avg R asks "how much do I make when I average across every outcome, right and wrong?" A strategy's real value comes from Avg R; Win % mainly affects how the experience of trading it *feels* day to day (a higher win rate makes for fewer, easier-to-tolerate losing streaks).

Avg PnL%

The plain underlying price move, averaged across all trades — e.g. "+0.055%" means the stock moved an average of 0.055% in the trade's favor. This is **not** capital return; it doesn't account for margin. At 4x MIS leverage, actual capital return is roughly 4× this figure (see the worked example below).

Trades

Sample size. Results based on fewer than ~100-200 trades should be treated cautiously — this research found multiple cases where a strategy looked excellent on a small sample and reverted much closer to average once tested on a larger one (see the caveats section).

2. Monthly profit calculation, ₹1,00,000 invested

Worked using **ma_alignment (confirmed)** — the strategy with the most consistent edge across this entire research effort — at 4x MIS margin, 0.5% stop. Two versions: the broad pooled average across all 165 stocks, and the single best-performing stock (IRCTC) specifically.

```
capital_return_per_trade = avg_underlying_pnl% × leverage(4x)
trades_per_month         = total_trades ÷ months_of_backtest (67 months)
monthly_return%          = trades_per_month × capital_return_per_trade
```

2a. Pooled average (all 165 stocks) — the realistic broad-based case

Metric	Value
Avg underlying PnL per trade (gross)	0.020%
Round-trip trading cost (brokerage + STT + exchange + GST + stamp duty, ₹4L notional)	0.047%
Avg underlying PnL per trade (net of costs)	**-0.027%**
Net capital return per trade (× 4 leverage)	**-0.108%**

The pooled, broad-based version of this strategy loses money after realistic trading costs. The raw backtest edge (+0.020% per trade) is smaller than the cost of simply placing the trade (0.047% round trip). This is the single most important finding in this whole report: a backtest showing a "positive" number is not sufficient — it has to be positive by more than the cost of trading it, and most of what was tested here is not.

2b. Best single stock — IRCTC specifically

Metric	Value
Trades (5-year backtest)	2,764
Avg underlying PnL per trade (gross)	0.055%
Win rate	43.2%

Round-trip trading cost	0.047%
Avg underlying PnL per trade (net of costs)	**0.008%**
Net capital return per trade (× 4 leverage)	**0.032%**
Net profit per trade, on ₹1,00,000	**₹32**
Trades per month (2,764 ÷ 67 months)	≈ 41.3
Net monthly profit, ₹1,00,000 invested	**≈ ₹1,320 (≈ 1.3%/month)**

Worth being direct about the trajectory of this number across the research: an earlier, smaller sample (196 trades over 6 months) suggested IRCTC's edge was worth roughly **15.8%/month gross**, before any cost adjustment. As the dataset grew to the full 5 years (2,764 trades), that same edge shrank to **0.055% gross** per trade — and after subtracting real trading costs, the honest net estimate is closer to **1.3%/month**. This is exactly the kind of correction more data is supposed to produce, and it lands close to the original 2-3%/month target rather than the inflated early estimate.

3. Top 10 stocks per strategy (5-year backtest, min. 100 trades)

All figures: 0.5% stop, arms at 1.5% (3x risk), give-back trail 0.5% behind peak — same exit rule throughout, so differences reflect entry logic only.

ma_alignment (confirmed) — overall best performer

#	Symbol	Trades	Avg R	Win %
1	LICI	1,945	+0.161	45.2%
2	BRITANNIA	2,392	+0.026	44.6%
3	APOLLOHOSP	2,535	+0.086	44.2%
4	ITC	2,267	+0.033	44.2%
5	AXISBANK	2,489	+0.083	43.9%
6	HINDUNILVR	2,270	+0.037	43.9%
7	HONAUT	2,489	+0.108	43.9%
8	BAJAJ-AUTO	2,440	+0.045	43.5%
9	ASIANPAINT	2,360	+0.033	43.4%
10	DMART	2,569	+0.110	43.4%

ma_alignment (4-tier, unconfirmed)

#	Symbol	Trades	Avg R	Win %
1	HINDUNILVR	2,290	+0.042	44.7%
2	LICI	1,949	+0.186	44.6%
3	ITC	2,301	+0.047	44.5%
4	DMART	2,626	+0.135	43.9%
5	BRITANNIA	2,479	+0.011	43.8%
6	APOLLOHOSP	2,562	+0.087	43.7%
7	AXISBANK	2,560	+0.060	43.3%
8	BAJAJ-AUTO	2,487	+0.060	43.3%
9	ASIANPAINT	2,430	+0.012	43.2%
10	LT	2,450	+0.059	43.2%

ma_alignment (3-tier, 45/30/15 — no 60m leg)

#	Symbol	Trades	Avg R	Win %
1	DMART	2,855	+0.150	44.2%
2	ITC	2,449	+0.063	43.8%
3	HINDUNILVR	2,461	+0.036	43.4%
4	LICI	2,131	+0.142	43.4%
5	MARUTI	2,651	+0.048	43.4%
6	BRITANNIA	2,640	+0.020	43.3%
7	HDFCBANK	2,377	+0.043	43.2%
8	LT	2,663	+0.043	42.7%
9	DRREDDY	2,685	-0.002	42.3%
10	DABUR	2,651	+0.019	42.2%

trend_confluence (vwap_trend + ma_alignment combined)

#	Symbol	Trades	Avg R	Win %
1	IRCTC	382	+0.155	49.2%
2	AWL	358	+0.212	48.9%
3	BERGEPAINT	442	+0.096	48.2%
4	LICI	347	+0.176	48.1%
5	CIPLA	363	+0.060	47.7%
6	AXISBANK	333	+0.034	47.1%
7	BAJAJ-AUTO	361	+0.099	47.1%
8	ADANIENT	330	+0.281	47.0%
9	DRREDDY	367	-0.014	46.6%
10	MARICO	393	-0.039	46.6%

vwap_trend (short-only, ≥45min into window)

#	Symbol	Trades	Avg R	Win %
1	LICI	313	+0.254	55.3%
2	AWL	347	+0.247	50.7%
3	ITC	350	+0.199	50.6%
4	3MINDIA	535	+0.148	47.3%
5	HDFCLIFE	343	+0.187	47.2%
6	EXIDEIND	396	+0.210	47.0%
7	COALINDIA	372	+0.210	46.5%
8	HONAUT	480	+0.077	46.3%
9	BAJAJ-AUTO	390	+0.136	46.2%
10	DIVISLAB	375	+0.058	45.3%

vwap_trend (unfiltered)

#	Symbol	Trades	Avg R	Win %
1	ITC	857	+0.053	44.8%
2	INFY	827	+0.023	42.4%
3	BAJAJ-AUTO	933	+0.058	42.2%
4	NESTLEIND	967	+0.010	42.2%
5	ASIANPAINT	908	-0.003	42.0%
6	HDFCBANK	861	+0.031	42.0%
7	APOLLOHOSP	956	+0.050	41.3%
8	DRREDDY	932	-0.046	41.3%
9	HCLTECH	946	+0.022	41.2%
10	HEROMOTOCO	962	+0.109	41.2%

trend_pullback

#	Symbol	Trades	Avg R	Win %
1	LT	2,407	+0.033	45.7%
2	INFY	2,259	+0.004	45.2%
3	MARUTI	2,369	+0.040	44.7%
4	M&M	2,548	+0.050	44.6%
5	HCLTECH	2,445	-0.007	44.2%
6	BRITANNIA	2,466	+0.005	44.0%
7	ASIANPAINT	2,361	0.000	43.9%
8	HDFCLIFE	2,522	+0.039	43.7%
9	BHARTIARTL	2,352	-0.001	43.6%
10	DABUR	2,443	+0.013	43.6%

momentum_breakout

#	Symbol	Trades	Avg R	Win %
1	AXISBANK	1,701	+0.045	45.4%
2	ASIANPAINT	1,758	-0.021	45.3%
3	LT	1,664	+0.051	45.3%
4	MARUTI	1,788	+0.038	45.2%
5	BRITANNIA	1,908	+0.006	44.7%
6	HCLTECH	1,735	-0.011	44.6%
7	INFY	1,585	0.000	44.5%
8	DMART	2,114	+0.090	44.2%
9	EICHERMOT	1,978	-0.004	44.2%
10	DRREDDY	1,870	-0.015	43.9%

strong_rally (0.1% threshold)

#	Symbol	Trades	Avg R	Win %
1	HINDUNILVR	748	+0.118	43.9%
2	HDFCBANK	637	+0.095	42.7%
3	LICI	822	+0.102	41.6%
4	ICICIBANK	817	+0.027	40.4%
5	LT	982	+0.029	40.1%
6	M&M	1,298	+0.065	40.1%
7	INFY	866	+0.014	40.0%
8	MARUTI	899	+0.023	39.8%
9	AUROPHARMA	1,641	+0.107	39.5%
10	BPCL	1,423	+0.095	39.5%

vwap_reversion

#	Symbol	Trades	Avg R	Win %
1	HDFCBANK	1,339	-0.017	42.3%
2	MARUTI	1,383	+0.038	42.1%
3	ASIANPAINT	1,381	0.000	41.5%
4	LT	1,394	+0.046	41.2%
5	HINDUNILVR	1,368	-0.017	40.9%
6	ICICIBANK	1,358	-0.010	40.9%
7	BRITANNIA	1,401	+0.022	40.8%
8	DIVISLAB	1,432	+0.098	40.7%
9	CIPLA	1,392	-0.055	40.2%
10	DRREDDY	1,392	-0.039	40.2%

strong_rally (0.2% threshold)

#	Symbol	Trades	Avg R	Win %
1	HINDUNILVR	131	+0.329	42.7%
2	ASIANPAINT	182	+0.147	41.2%
3	ABBOTINDIA	235	+0.044	40.0%
4	HINDALCO	414	+0.108	39.9%
5	AUROPHARMA	495	+0.171	39.0%
6	ICICIBANK	136	-0.049	39.0%
7	HEROMOTOCO	320	+0.180	38.4%
8	DLF	530	+0.149	38.3%
9	M&M	347	+0.053	38.3%
10	DMART	308	+0.050	38.0%

orb

#	Symbol	Trades	Avg R	Win %
1	HDFCBANK	419	+0.113	44.4%
2	HINDUNILVR	484	-0.006	40.1%
3	LICI	420	+0.142	40.0%
4	ITC	473	+0.018	39.1%
5	MARUTI	527	+0.067	39.1%
6	NESTLEIND	448	-0.061	36.6%
7	COALINDIA	537	+0.058	36.5%
8	BAJFINANCE	494	+0.161	36.4%
9	LINDEINDIA	525	+0.161	36.4%
10	LT	531	-0.009	36.3%

Cross-cutting pattern: LICI, HINDUNILVR, ITC, BRITANNIA, AXISBANK, BAJAJ-AUTO and DMART appear repeatedly near the top across multiple independently-built strategies. That recurrence across different entry logics is more meaningful than any single strategy's top-10 list — it suggests these specific stocks have a genuine structural tendency to trend cleanly, rather than one strategy happening to fit them by chance.

4. Caveats — read before acting on any of this

- **Costs matter more than most of the edges found here.** Round-trip trading cost (~0.047% of notional at ₹4L) is larger than the pooled average edge of most strategies tested. Only strategy/stock combinations with a clearly larger gross edge survive contact with real costs.
- **Small samples inflate results.** This report only shows combinations with 100+ trades, but even those can look better than they'll perform going forward — several strategies produced dramatically different top performers when the dataset grew from 6 months to 5 years (e.g. DLF's `strong_rally` result fell from +1.115 avg R on 31 trades to +0.316 avg R once retested on 119).
- **No true out-of-sample test has been run yet.** All of this was backtested and then reported on the same data — the responsible next step is validating the top candidates on a period the strategy wasn't tuned against.
- **Paper-fill assumptions are optimistic.** Backtested fills assume execution at the exact modeled price; real slippage (confirmed once already this session, on an LT trade that overshot its stop within one bar) will be somewhat worse in practice.
- None of this has been validated with real capital. Treat every number here as a research finding, not a trading recommendation.

Generated from the papertrader research pipeline (SQLite-backed, `papertrader research` CLI) — full per-trade detail queryable via run ids 44-58 in `data/papertrader.db`. Source markdown reports: `research/fiveyear-v1.md`, `research/confluence-v1.md`.